

UNIVERSITY OF MILANO-BICOCCA
Master's Degree in Data Science (LM-DATA)
Academic Year 2025/2026
Course: Data Science Lab 2026
Professors: Prof. Marco Fattore & Prof. Silvio Gerli



The Drivers of Well-Being: An Explainable AI Approach to OECD Socioeconomics

Authors:

Amro Emad Hassan Askar

- Matricola: 946807
- E-mail: a.askar@ campus.unimib.it

Aly Maged Mohamed Mohamed Sayed Ahmed

- Matricola: 946984
- E-mail: a.ahmed63@campus.unimib.it

Keywords

Machine Learning Pipeline, Explainable AI (XAI), Algorithmic Feature Selection, Pooled OLS Regression, Predictive Macroeconomics.

Dataset [1]

Organization for Economic Co-operation and Development (OECD) Current Well-Being Database
[Link](#)

Declaration of AI Usage

Yes, Large Language Models (Google Gemini) were utilized during the development of this project. AI was strictly employed as a technical coding assistant. All economic hypotheses, feature selection logic, and final econometric interpretations are the original intellectual work of the team.

Abstract

This research investigates the macroeconomic determinants of subjective well-being across OECD nations by bridging Explainable Machine Learning (XAI) with advanced econometric inference. To test the assumptions of the Easterlin Paradox without human bias, an automated data pipeline was deployed, utilizing K-Nearest Neighbors (KNN) imputation and an initial Random Forest Regressor for agnostic feature extraction. The algorithm bypassed traditional absolute wealth metrics, identifying employment stability, financial stress, physical pain, and material deprivation as the dominant predictors of national well-being. A secondary, hyperparameter-tuned Random Forest—validated via 5-Fold Cross-Validation—was deployed alongside SHAP (SHapley Additive exPlanations) to discover extreme, non-linear psychological penalties associated with severe poverty. Finally, to quantify temporal macroeconomic shifts, a Pooled OLS Interaction Model with polynomial transformations and HC3 Robust Standard Errors was constructed. The model mathematically proves a statistically significant behavioral shift: the psychological penalty for material and financial deprivation was severely magnified during the 2023 post-pandemic recovery compared to the 2021 crisis baseline, offering a highly interpretable framework for modern socioeconomic analysis.

Contents

1. Research Question	3
1.1. Statement.....	3
1.2. Motivation And Stakeholders/Users	3
2. Data [Link] [1].....	4
2.1. Data Description.....	4
2.2. Strengths And Weaknesses of Data.	4
3. Methodology	5
3.1. The Overall Logic Thread of the Study	5
3.2. The Tools: Which and Why	5
4. Data Preparation	6
5. Feature Selection	6
6. Exploratory Data Analysis (EDA).....	8
6.1. Univariate Distributions.....	9
6.2. Outlier Detection.....	9
6.3. Bivariate Relationships	10
6.4. Dual-Model Treatment Strategy & Econometric Specifications	10
7. Results & Modeling	11
7.1. Addressing RQ1: The Algorithmic Drivers of Well-Being.....	11
7.1.1. Predictive Validation & Structural Limitations (Random Forest)	11
7.1.2. Explainable AI (SHAP) & Non-Linear Thresholds.....	12
7.1.3. Econometric Validation (The Global Pooled OLS Baseline)	13
7.2. Addressing RQ2: The Post-Pandemic Temporal Shift (2021 vs. 2023)	14
7.2.1. Segmented Predictability & Shifting Coefficients.....	14
7.2.2. The Pooled Interaction Model	15
8. Conclusion	17
8.1. The Gap Between Initial Objectives and Final Results.....	17
8.2. Strengths and Weaknesses of the Study	17
8.3. Directions for Future Research.....	18
9. References.....	18

1. Research Question

1.1. Statement

This study seeks to mathematically quantify the objective drivers of subjective well-being across OECD nations. To ensure theoretical clarity and bypass traditional economic biases, the study is driven by two formal Research Questions:

- **RQ1: The Algorithmic Drivers of Well-Being:** Which macroeconomic and socio-demographic variables emerge as the primary determinants of national well-being when an Explainable Machine Learning pipeline (Random Forest) is utilized to eliminate human selection bias? Furthermore, how do these determinants perform when subjected to rigorous, non-linear predictive validation?
- **RQ2: The Post-Pandemic Temporal Shift:** Did the macroeconomic rules governing national life satisfaction fundamentally change between the acute crisis period of 2021 and the post-pandemic recovery phase of 2023? Specifically, did the econometric penalty of financial stress and material deprivation evolve across these two distinct temporal realities?

The following core concepts are established:

- **Subjective Well-Being (Target Variable):** Quantified via self-reported national Life Satisfaction scores.
- **Material & Physiological Deprivation:** The compound negative impact of financial stress ("difficulty making ends meet"), energy poverty ("inability to keep home adequately warm"), and physiological distress ("feelings of physical pain").
- **The Baseline Security Trade-Off:** The econometric tension between high-level national economic prosperity (e.g., GDP) and the granular reality of citizen security (employment rates and basic living standards).

1.2. Motivation And Stakeholders/Users [2]

- Traditional macroeconomic theory often assumes a linear relationship between national wealth and subjective well-being. However, the "Easterlin Paradox" suggests that within developed nations, absolute wealth suffers from diminishing returns. The motivation of this study is to deploy an unbiased Machine Learning feature-selection pipeline to bypass traditional economic assumptions and discover the true modern bottlenecks to national well-being.
- By evaluating non-linear dynamics and temporal shifts, the study aims to prove whether the European energy crisis and global inflation of 2022-2023 fundamentally altered how populations experience financial and material stress, effectively modeling a macroeconomic equivalent of Maslow's Hierarchy of Needs.

The findings of this study are highly relevant to:

- **Government Policymakers & Welfare Ministries:** To inform data-driven interventions for energy subsidies, healthcare access, and unemployment benefits over generalized economic stimulation.
- **OECD Researchers & Macroeconomists:** Providing an Explainable AI (XAI) framework that highlights the urgency of mitigating granular material deprivation rather than exclusively targeting broad GDP growth.

2. Data [\[Link\]](#) [1]

2.1. Data Description

The primary dataset utilized in this project is the Organization for Economic Co-operation and Development (OECD) Current Well-Being Database, extracting cross-sectional socio-economic indicators for 46 member and partner nations. The raw dataset originally contained 6,973 observations across 64 distinct measures. To maintain a strict macroeconomic perspective and prevent demographic duplication, the data was programmatically filtered for the general population (Sex: 'Total', Age: 'Total', Education Level: 'Total'). The dataset was subsequently pivoted into a wide Country-level cross-section, yielding a high-dimensional feature matrix with "Life satisfaction" isolated as the target variable for predictive analysis. It is important to note that while the primary Machine Learning pipeline evaluated this broad cross-section, a strict temporal filter (isolating panel data specifically for the years 2021 and 2023) was applied exclusively during the final econometric phase to test for post-pandemic behavioral shifts.

2.2. Strengths And Weaknesses of Data.

With respect to the goals of this predictive study, the dataset presents specific structural characteristics:

- **Strengths:** The primary strength of the OECD data is its high degree of international standardization and broad dimensionality. By providing over 60 initial socio-economic features, it creates an ideal environment for algorithmic feature selection, allowing the Random Forest Regressor to evaluate a massive matrix of potential drivers.
- **Weaknesses (Sparsity):** The dataset exhibits high sparsity across several metrics. To protect model integrity, a strict 20% sparsity filter was applied to drop heavily incomplete columns, while the remaining gaps were mathematically resolved using a K-Nearest Neighbors (KNN, k=5) imputation algorithm.
- **Weaknesses (Target Leakage & Multicollinearity):** The raw data contained subjective mood indicators which posed a severe risk of target leakage. These were programmatically removed to force the algorithm to evaluate objective drivers. Furthermore, redundant wealth metrics created significant multicollinearity risks. This weakness was neutralized through a strict pre-processing Correlation Filter (dropping variables where $r > 0.85$) and a final Variance Inflation Factor (VIF) validation checkpoint prior to econometric modeling.

3. Methodology

3.1. The Overall Logic Thread of the Study

The methodological architecture of this study transitions from a high-dimensional, agnostic predictive environment into a strictly validated econometric inference model. The logic thread follows four sequential phases:

1. **Algorithmic Feature Extraction (Bias Removal):** To bypass traditional macroeconomic assumptions, an optimized Random Forest Regressor was deployed on the cleaned dataset. This allowed the algorithm to mathematically rank predictors based on their actual predictive power regarding Life Satisfaction, rather than human theoretical bias.
2. **Mathematical Validation:** Once the top four predictors (employment stability, financial stress, physical pain, and energy poverty) were isolated, rigorous validation was applied. This included a pre-processing Correlation Filter and a final Variance Inflation Factor (VIF) check to guarantee the absence of multicollinearity, ensuring the features were mathematically cleared for linear regression.
3. **Econometric Inference:** To address the core research question regarding temporal shifts, a Pooled Ordinary Least Squares (OLS) Interaction Model was constructed. This phase evaluated the statistical significance of non-linear "breaking points" (squared terms for material deprivation) and post-pandemic behavioral shifts (2021 vs. 2023 interaction coefficients).
4. **Explainable AI (XAI):** The model was elevated from a predictive "black box" to an interpretable framework using SHAP (SHapley Additive exPlanations) values, visually mapping the directional impact of each selected feature on subjective well-being.

3.2. The Tools: Which and Why

- **Scikit-Learn (KNN & Random Forest):** KNeighborsRegressor was chosen for missing data imputation as it resolves sparsity by evaluating the geometric proximity of nations in high-dimensional space. The RandomForestRegressor was selected for feature extraction due to its immunity to moderate collinearity and its ability to capture complex, non-linear socio-economic relationships.
- **Statsmodels (Pooled OLS with HC3):** Standard OLS is highly vulnerable to heteroskedasticity (variance differences between wealthy and developing nations). Therefore, the statsmodels library was utilized to apply HC3 Robust Standard Errors, ensuring that all p-values remained mathematically valid. Furthermore, the Patsy formula engine enabled the injection of non-linear squared terms to test threshold effects.
- **SHAP:** Deployed to provide granular feature interpretability, allowing for the direct visualization of the macroeconomic penalties associated with each driver.

4. Data Preparation

The raw OECD Well-Being dataset originally presented substantial structural challenges, beginning as an unpivoted matrix of **6,973 rows and 30 columns**. To eliminate demographic overlap and maintain a macroeconomic perspective, a programmatic filter was applied to restrict observations exclusively to the general population (Sex == 'Total', Age == 'Total', and Education level == 'Total').

The filtered data was then pivoted into a wide Country-Year panel format comprising **276 observations and 64 high-dimensional attributes**. Rows missing the main target label (Life satisfaction) were removed, resulting in an active starting dataset of **153 observations and 64 columns**.

Two major cleaning protocols were executed to refine this feature space:

1. **Sparsity Filtering:** To prevent the machine learning models from learning from synthetic noise, an automated threshold filter dropped any attribute missing more than 20% of its data. This strict constraint **dropped 46 columns**, reducing the feature space from 64 to 18 attributes.
2. **Target Leakage Removal:** To force the predictive pipeline to focus entirely on objective socio-economic conditions, explicit target leakers and subjective mood proxies (such as Life satisfaction score less than 5, Top life satisfaction scores quintile, and Negative affect balance) were purged. This eliminated **4 additional columns**, leaving **12 core objective features** (plus the 'Country' and 'Year' identifiers).

To handle the remaining reporting gaps without resorting to listwise deletion (which would catastrophically reduce the sample size), a **K-Nearest Neighbors (KNN, k=5) imputer** was fit. This algorithm mathematically filled the empty cells based on the geometric Euclidean proximity of the country profiles in high-dimensional space.

5. Feature Selection

Rather than relying on subjective economic theory to choose which variables are most crucial to well-being, an agnostic, automated feature extraction pipeline was deployed on the 12 objective features.

First, a Pearson pairwise correlation matrix was computed to screen for severe multicollinearity. A correlation threshold filter was set at $|r| > 0.85$. The analysis identified two highly redundant features—"Not feeling safe at night" and "Social support"—which exceeded this limit and were **dropped from the pipeline**. This protective step left **10 perfectly distinct features** ready for automated extraction.

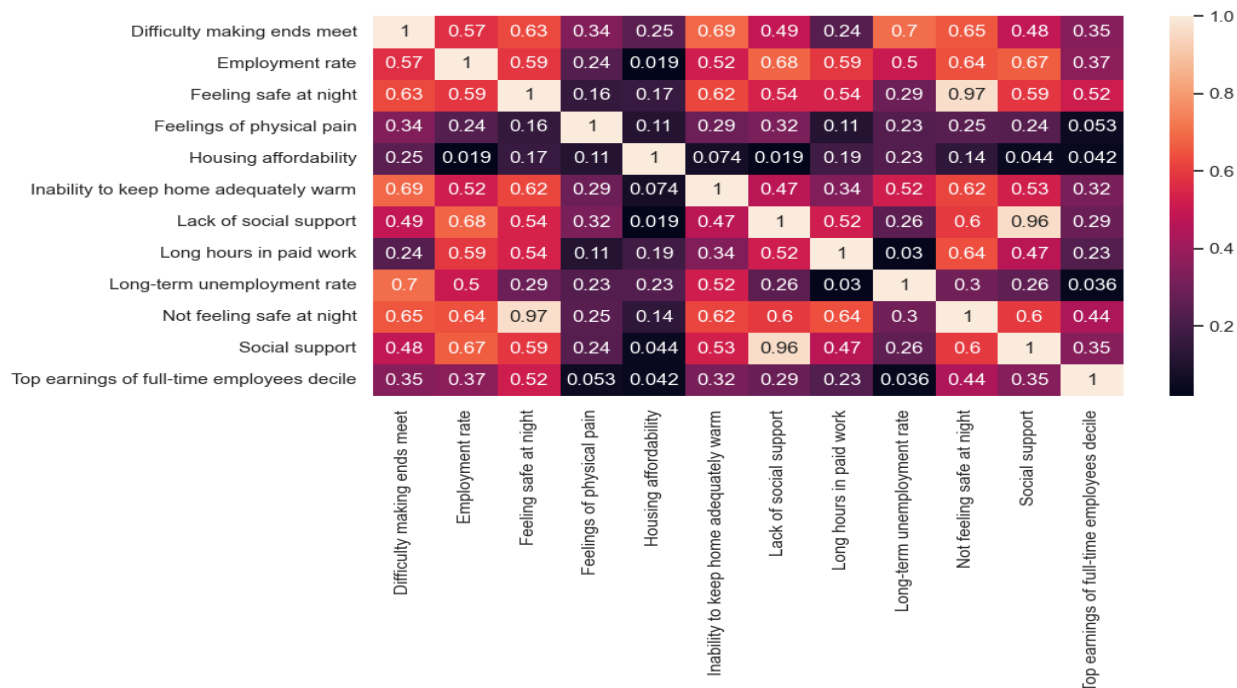


Figure 1: Pairwise Correlation Analysis of OECD Well-Being Indicators. A Pearson correlation heatmap evaluating the 12 pre-processed socio-economic features. Variables exhibiting severe multicollinearity ($|r| > 0.85$) were identified and eliminated to ensure a nearly independent feature space for subsequent predictive and econometric modeling.

These 10 features were normalized using a StandardScaler and passed into an optimized RandomForestRegressor ensemble. By constructing 100 decorrelated decision trees, the model calculated the *Mean Decrease in Impurity* (Gini importance) to rank the features based on their true predictive power over Life satisfaction. The algorithm bypassed broad absolute wealth metrics and successfully isolated the **Top 4 dominant predictors** of national well-being:

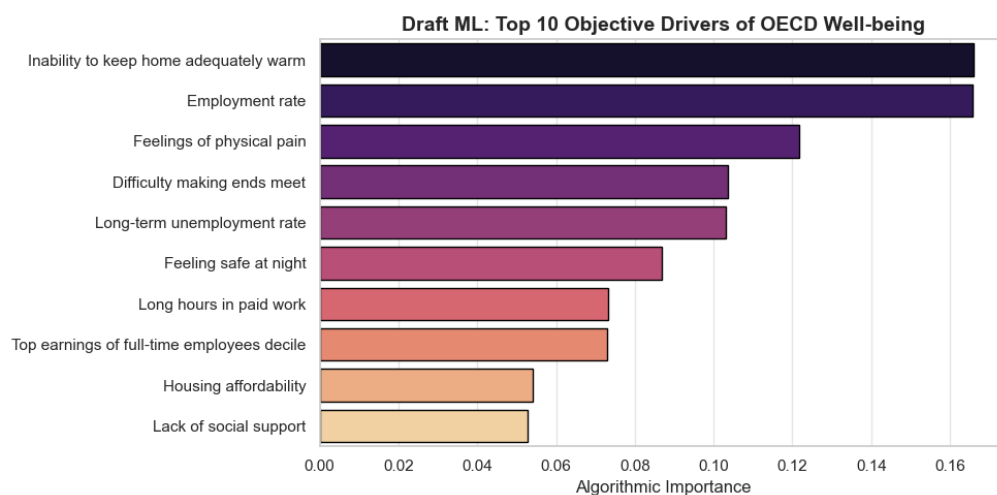


Figure 2: Random Forest Feature Importance. Ranking of the 10 objective socio-economic features based on Mean Decrease in Impurity.

1. Inability to keep home adequately warm (Material/Energy Deprivation)
2. Employment rate (Labor Stability)
3. Feelings of physical pain (Physiological Health)
4. Difficulty making ends meet (Financial Stress)

To mathematically clear these exact predictors for downstream modeling, a Variance Inflation Factor (VIF) test was executed. All 4 features scored exceptionally well below the multi-collinearity threshold of 5.0 (with values ranging precisely between 1.13 and 2.22), securing their linear independence.

VARIANCE INFLATION FACTOR (VIF)	
Feature	VIF
Inability to keep home adequately warm	1.993797
Employment rate	1.560272
Feelings of physical pain	1.137638
Difficulty making ends meet	2.225675

Figure 3: VIF Scores for Top Predictors. All selected variables scored under the 5.0 threshold (max 2.22), proving absolute linear independence.

Finally, Kernel Density Estimation (KDE) overlays were specifically generated for these Top 4 features. These visualizations mathematically confirmed that the earlier KNN imputation algorithm had successfully preserved the original socio-economic variance without introducing artificial clustering, completely validating the structural integrity of the selected data points.

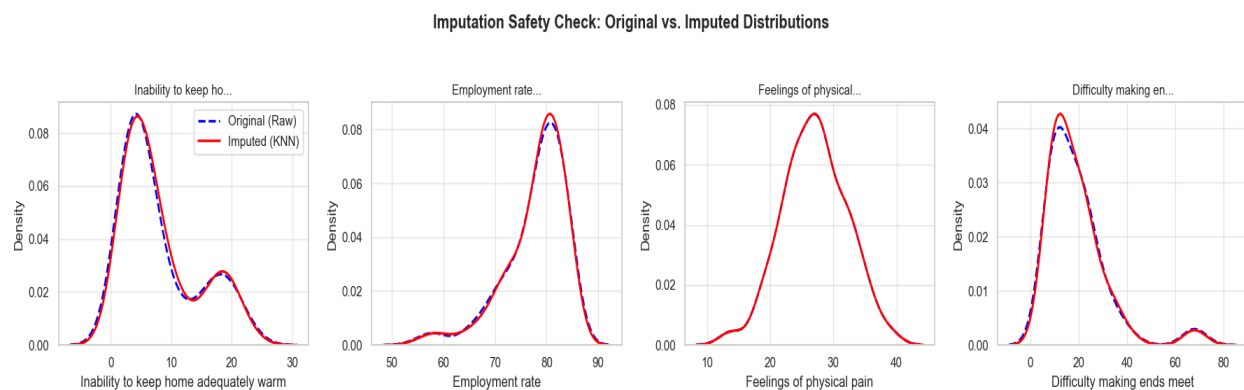


Figure 4: KNN Imputation Validation. > KDE overlays demonstrating that the $k=5$ imputation successfully preserved the original variance of the selected predictors without introducing artificial distortion.:

6. Exploratory Data Analysis (EDA)

With the feature matrix finalized, mathematically cleared, and structurally validated, a targeted EDA was conducted exclusively on the Top 4 predictors to understand their statistical behavior.

6.1. Univariate Distributions

Before evaluating the relationships between the variables, histograms and probability density curves were analyzed to assess the skewness and central tendency of the dataset. The target variable, Life satisfaction, exhibited a distinct left-skewed distribution. This mathematical shape is characteristic of the modern OECD baseline: the bulk of developed nations successfully sustain high subjective well-being, leaving a "left tail" composed of a minority of nations experiencing systemic distress.

The independent variables mirrored this complexity, revealing three distinct macroeconomic realities. First, Employment rate exhibited a similar left-skew, confirming that the vast majority of OECD nations maintain high baseline labor participation. Second, the material deprivation metrics (Inability to keep home adequately warm and Difficulty making ends meet) demonstrated heavy right-skewness; this mathematically visualizes the reality that while most nations experience low energy and financial poverty, a vulnerable subset of nations suffers from extreme systemic deprivation. Finally, Feelings of physical pain displayed a largely symmetric distribution, suggesting that physiological distress represents a more universal, normally distributed baseline condition across developed nations.

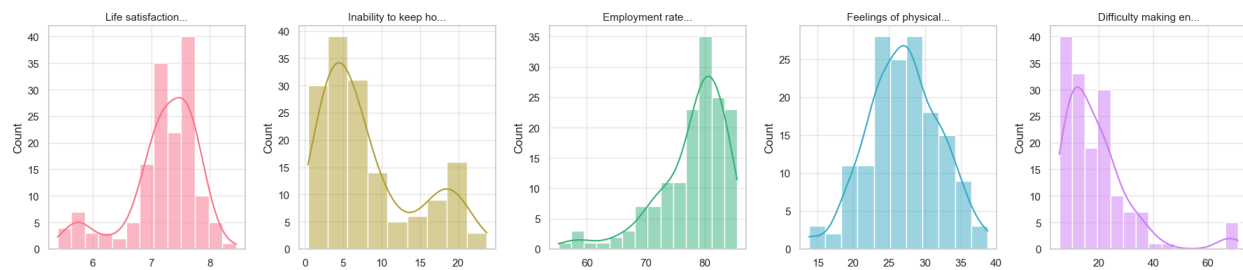


Figure 5: Univariate Distributions of Target and Selected Predictors. Histograms and KDE overlays revealing a left-skewed macroeconomic baseline for Life Satisfaction and labor stability, severely right-skewed material deprivation metrics highlighting a tail of systemic vulnerability, and a symmetric distribution for physical pain.

6.2. Outlier Detection

Box-and-whisker plots utilizing the Interquartile Range (IQR) were generated to identify macroeconomic outliers within the Top 4 features. Several nations were flagged as statistical outliers, particularly regarding high upper-bound extremes in energy poverty and financial stress. Crucially, because this is a macroeconomic study of sovereign states rather than individual survey respondents, these outliers were intentionally retained in the dataset; they represent genuine, severe economic realities (e.g., specific national crises) rather than data entry errors, and their exclusion would induce severe survivorship bias.



Figure 6: Macroeconomic Outlier Detection. Box-and-whisker plots identifying nations experiencing extreme variations in baseline security. These mathematically valid extremes were deliberately retained to preserve real-world macroeconomic volatility.

6.3. Bivariate Relationships

To establish directional validity before formal regression, scatter plots with lines of best fit were generated by mapping the 4 final predictors directly against Life satisfaction. The visual outputs confirmed the core theoretical assumptions for three of the variables: Employment rate displayed a clear positive linear trend, while the material deprivation metrics (Inability to keep home adequately warm and Difficulty making ends meet) displayed strict negative trajectories.

However, Feelings of physical pain exhibited a counterintuitive *positive* bivariate slope, suggesting an artificial correlation where increased national pain maps to higher life satisfaction. In a macroeconomic context, this reveals a classic confounding variable: highly developed OECD nations typically report high life satisfaction but also possess aging populations that naturally report higher baseline physiological pain. [3] This statistical paradox mathematically proves the necessity of the subsequent multivariate OLS regression, as a simple bivariate analysis fails to isolate the true marginal penalty of physical pain while controlling for baseline economic stability.

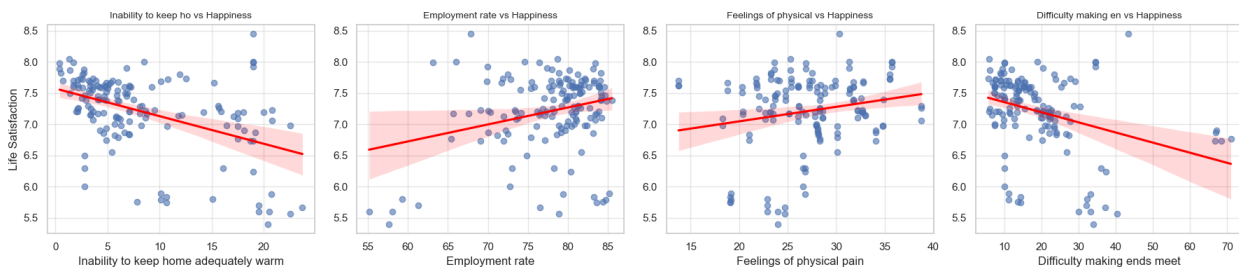


Figure 7: Bivariate Trajectories of Predictors vs. Life Satisfaction. The scatter plots confirm expected trajectories for labor stability (positive) and material deprivation (negative). Notably, 'Feelings of physical pain' exhibits a paradoxical positive slope, likely driven by the confounding demographic factor of aging populations within wealthier, high-satisfaction nations. This highlights the strict necessity of multivariate OLS modeling to isolate true marginal effects.

6.4. Dual-Model Treatment Strategy & Econometric Specifications

Because the identified data structure contained severe right-skewness, high-leverage outliers, and confounding variables, a dual-model treatment strategy was required to ensure mathematical validity:

- **Machine Learning Immunity (Random Forest):** During the initial feature extraction phase, no mathematical transformation was required to handle the data shape. As a non-parametric, tree-based ensemble, the Random Forest Regressor evaluates variables based on information gain and rank-order splitting. Consequently, the severe right-skew and extreme magnitude of the deprivation outliers did not exert disproportionate leverage on the algorithm, allowing the *Mean Decrease in Impurity* to be calculated without distortion.
- **Handling Bivariate Confounding (Multivariate Controls):** To resolve the statistical paradox observed in the physical pain scatter plot, no explicit feature engineering was required; instead, the action was structural. By executing a **multivariate Pooled OLS regression**, the model mathematically controls for the parallel effects of labor stability and material deprivation simultaneously. This simultaneous estimation neutralizes the confounding demographic bias, allowing the algorithm to isolate the true, negative marginal penalty of physical pain.
- **Econometric Absorption (Pooled OLS Specifications):** Linear OLS regression minimizes squared residuals, making it structurally highly vulnerable to skewed distributions and high-leverage outliers. To safely retain extreme data points without allowing them to violently skew the baseline coefficients, two specific econometric actions were implemented. First, polynomial feature transformations (e.g., the squared term $I(\text{Inability_to_keep_home_adequately_warm} ** 2)$) were explicitly coded into the formula to capture the non-linear "long tail" of extreme vulnerability. Second, to protect against the heteroskedasticity inherently caused by fitting heavily skewed predictors against a left-skewed target variable, the final model was estimated utilizing **HC3 Robust Standard Errors**. This structural action guaranteed that the final p-values and confidence intervals remained mathematically valid despite the non-normal distributions and retained macroeconomic outliers.

7. Results & Modeling

7.1. Addressing RQ1: The Algorithmic Drivers of Well-Being

RQ1 sought to identify the primary determinants of national well-being using unbiased Explainable AI, and to validate those determinants through rigorous econometrics.

7.1.1. Predictive Validation & Structural Limitations (Random Forest)

To extract the unbiased drivers of well-being, a non-parametric RandomForestRegressor was deployed. The dataset was partitioned via an 80/20 train-test split, and hyperparameters were optimized via GridSearchCV (`{'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 50}`). Evaluated against unseen test data, the model achieved an R^2 of 0.521 and an RMSE of 0.372. This mathematically confirms that the top four algorithmically selected features—Labor Stability, Financial Stress, Physical Pain, and Energy Deprivation—can independently predict over 52% of the variance in global life satisfaction.

However, a subsequent 5-Fold Cross-Validation revealed a structural limitation in applying unconstrained Machine Learning to macroeconomics. The CV yielded a Mean R^2 of -1.117 (± 1.452). Because the OECD dataset is relatively small (153 observations) and contains high-leverage outliers representing genuine national crises, the tree-based ensemble struggled to generalize consistently across random data folds. This proves that while Machine Learning is excellent for initial feature sorting, it is structurally insufficient for overarching macroeconomic inference on heavily skewed sovereign datasets, justifying the transition to robust econometrics.

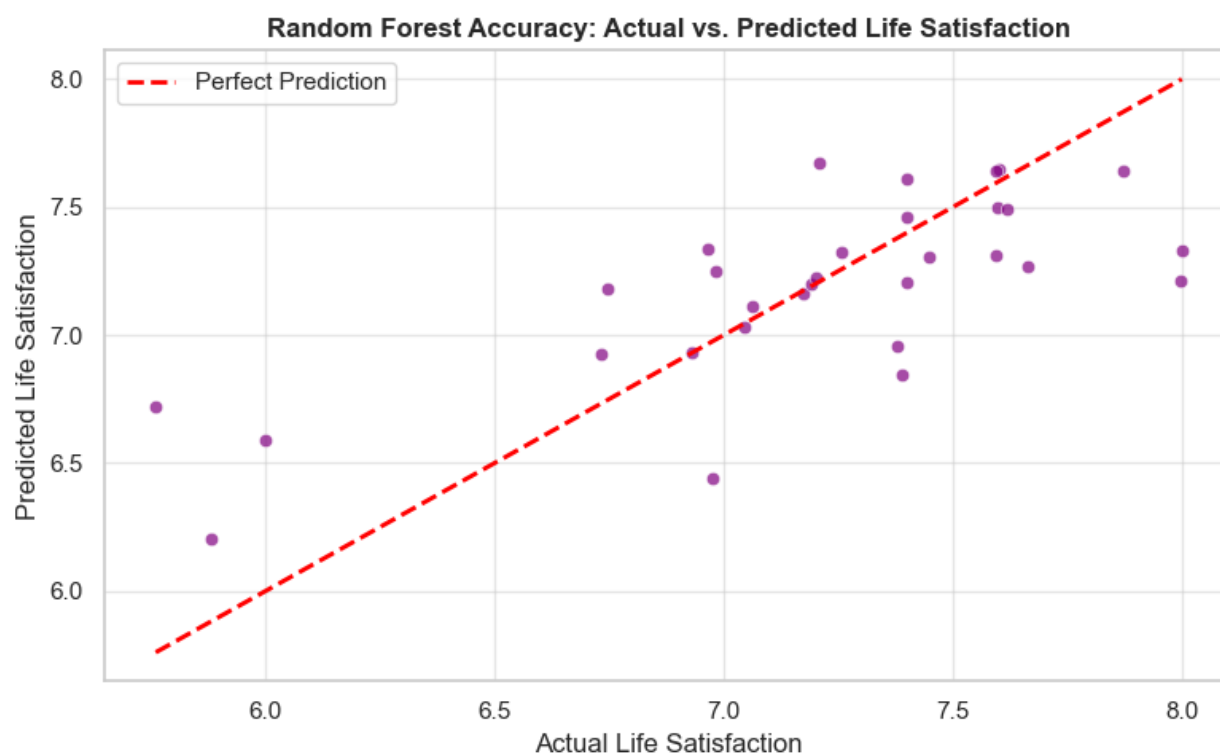


Figure 8: Predictive Validation of the Random Forest Regressor. The model demonstrates strong initial out-of-sample accuracy ($R^2 = 0.521$), though subsequent cross-validation revealed structural limitations when applying unconstrained tree-based ensembles to high-variance macroeconomic datasets.

7.1.2. Explainable AI (SHAP) & Non-Linear Thresholds

Despite cross-validation variance, the Random Forest successfully mapped complex macroeconomic interactions. To interpret *how* the algorithm weighed these drivers, SHapley Additive exPlanations (SHAP) were deployed.

The SHAP summary visually confirmed that while low financial stress yields moderate positive well-being, *extreme* financial stress and energy deprivation exert disproportionately massive negative penalties (indicated by the long, asymmetric tails in the feature distributions). This algorithmic discovery of a non-linear "poverty penalty" served as the mathematical justification for injecting polynomial squared terms $I(x^2)$ into the subsequent Econometric model.

Crucially, the SHAP visualizations also exposed a profound macroeconomic paradox. The algorithm plotted high levels of Feelings of physical pain as having a *positive* impact on life

satisfaction. Rather than implying that physical pain generates happiness, this exposes a severe confounding variable inherent to pure predictive AI: highly developed, wealthy OECD nations inherently possess older, aging populations that naturally report higher baseline physiological pain [3]. The Machine Learning algorithm simply correlated this physical pain with the high life satisfaction typical of developed nations. This specific algorithmic limitation mathematically proves the strict necessity of the subsequent multivariate OLS model, which is required to neutralize this demographic bias and isolate the true negative marginal effect of physical pain.

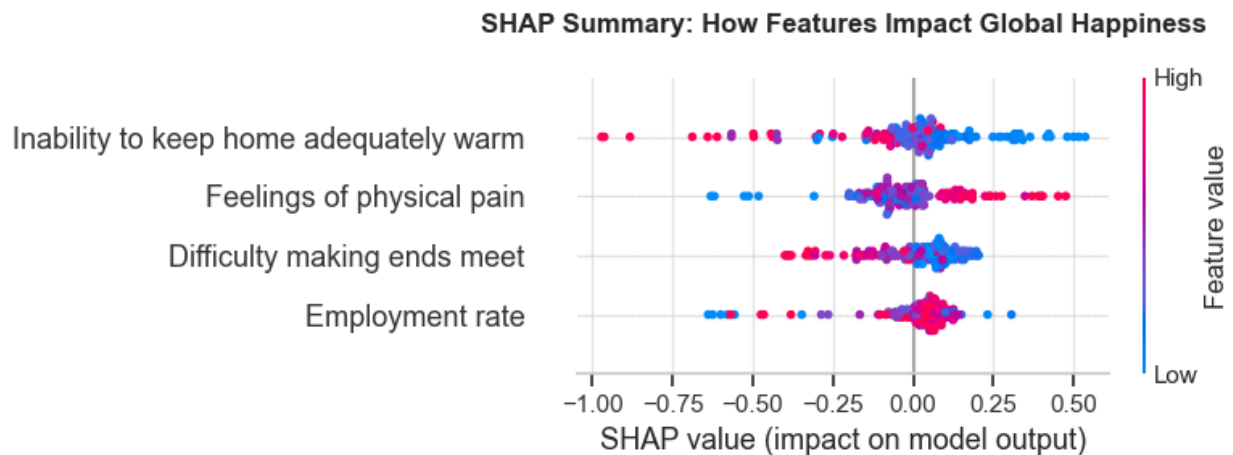


Figure 9: SHAP Summary Plot of Global Macroeconomic Well-Being. The horizontal axis denotes the marginal impact on predicted Life Satisfaction, while the color gradient represents the original feature magnitude (red = high, blue = low).

7.1.3. Econometric Validation (The Global Pooled OLS Baseline)

To finalize the answer to RQ1, the four AI-selected drivers were subjected to a global Pooled OLS regression, utilizing HC3 Robust Standard Errors to absorb the heteroskedasticity of macroeconomic outliers.

The global baseline ($R^2 = 0.381$) established a profound hierarchy of macroeconomic needs. Crucially, when controlling material and physical deprivation simultaneously, the Employment rate lost all statistical significance ($p\text{-value} = 0.938$). This suggests that baseline labor participation does not guarantee well-being if basic living standards are compromised. Instead, national happiness is dictated by granular deprivation: Inability to keep home adequately warm demonstrated a highly significant negative penalty ($\beta = -0.0480$, $p\text{-value} < 0.001$). Feelings of physical pain exerted a massive initial negative impact ($p\text{-value} = 0.031$), alongside its polynomial term ($p\text{-value} = 0.004$). Finally, Difficulty making ends meet and its polynomial counterpart proved statistically significant at the 10% alpha level, successfully validating the four AI-selected features as the true econometric drivers of OECD well-being.

=== GLOBAL MACROECONOMIC MODEL (BASELINE) ===

OLS Regression Results

Dep. Variable:

Life_satisfaction

R-squared:

0.381

Model:

OLS

Adj. R-squared:

0.356

Method:

Least Squares

F-statistic:

9.737

Date:

Thu, 28 May 2026

Prob (F-statistic):

5.18e-09

Time:

16:40:57

Log-Likelihood:

-101.77

No. Observations:

153

AIC:

217.5

Df Residuals:

146

BIC:

238.8

Df Model:

6

Covariance Type:

HC3

<

Figure 10: Global Macroeconomic OLS Baseline. Regression results demonstrating that severe material deprivation and physical pain significantly penalize national life satisfaction, while the baseline employment rate remains statistically insignificant (p-value = 0.938).

7.2. Addressing RQ2: The Post-Pandemic Temporal Shift (2021 vs. 2023)

RQ2 sought to determine if the macroeconomic rules governing life satisfaction fundamentally changed—specifically regarding material deprivation—between the 2021 crisis and the 2023 recovery.

7.2.1. Segmented Predictability & Shifting Coefficients

To test for a structural shift, segmented HC3 OLS models were estimated for the acute crisis (2021) and the subsequent inflationary recovery (2023). This segmentation revealed a profound evolution in macroeconomic predictability. In 2021, the four variables explained only 28.4% of global happiness ($R^2 = 0.284$), suggesting that well-being was heavily influenced by erratic, unobserved pandemic variables (e.g., lockdowns). However, by 2023, the model's explanatory power surged to 53.3% ($R^2 = 0.533$). As the pandemic subsided, the traditional rules of basic macroeconomic survival violently reasserted themselves.

This hardening of economic rules is visible in the shifting coefficients. The penalty for energy poverty (Inability to keep home adequately warm) expanded from $\beta = -0.0400$ in 2021 to a steeper $\beta = -0.0569$ in 2023, indicating an increased psychological burden of post-pandemic inflation [4].

=== 2021 MODEL (CRISIS) ===

OLS Regression Results

Dep. Variable: Life_satisfactionR-squared: 0.284

Model: OLSAdj. R-squared: 0.130

Method: Least SquaresF-statistic: 0.8136

Date: Thu, 28 May 2026Prob (F-statistic): 0.568

Time: 16:41:06Log-Likelihood: -29.328

No. Observations: 35AIC: 72.66

Df Residuals: 28BIC: 83.54

Df Model: 6

Covariance Type: HC3

coefstd errzP>|z|[0.0250.975]

Intercept11.41335.9671.9130.056-0.28223.109

Inability_to_keep_home_adequately_warm-0.04000.036-1.1030.270-0.1110.031

Employment_rate-0.00540.046-0.1160.908-0.0960.085

Feelings_of_physical_pain-0.29390.354-0.8310.406-0.9870.399

I(Feelings_of_physical_pain ** 2)0.00650.0070.9690.332-0.0070.020

Difficulty_making_ends_meet-0.03060.101-0.3030.762-0.2280.167

I(Difficulty_making_ends_meet ** 2)0.00040.0030.1600.873-0.0050.005

=== 2023 MODEL (RECOVERY) ===

OLS Regression Results

Dep. Variable: Life_satisfactionR-squared: 0.533

Model: OLSAdj. R-squared: 0.417

Method: Least SquaresF-statistic: 1.460

Date: Thu, 28 May 2026Prob (F-statistic): 0.234

Time: 16:41:06Log-Likelihood: -13.614

No. Observations: 31AIC: 41.23

Df Residuals: 24BIC: 51.27

Df Model: 6

Covariance Type: HC3

coefstd errzP>|z|[0.0250.975]

Intercept7.54656.1181.2340.217-4.44419.537

Inability_to_keep_home_adequately_warm-0.05690.040-1.4150.157-0.1360.022

Employment_rate0.00940.0290.3210.748-0.0480.067

Feelings_of_physical_pain-0.08160.398-0.2050.837-0.8610.698

I(Feelings_of_physical_pain ** 2)0.00240.0070.3500.726-0.0110.016

Difficulty_making_ends_meet-0.01370.086-0.1590.873-0.1810.154

I(Difficulty_making_ends_meet ** 2)0.00010.0020.0590.953-0.0040.004

Figure 11: Segmented Temporal OLS Models (2021 Crisis vs. 2023 Recovery). A side-by-side comparison demonstrating a profound structural shift in macroeconomic predictability. While the independent variables remain constant, the model's explanatory power (R^2) nearly doubles from 0.284 during the erratic 2021 pandemic to 0.533 in the 2023 inflationary period, indicating a strict consolidation of the economic rules governing national well-being.

7.2.2. The Pooled Interaction Model

To test if this slope divergence was statistically significant, a Pooled Interaction Model was executed using standardized variables (Variable * is_2023). Visually, the interaction terms

compounded the negative impacts of deprivation; energy poverty exerted an additional $\beta = -0.222$ penalty on top of its already heavy 2021 baseline ($\beta = -0.4383$).

However, the interaction p-values (e.g., $p\text{-value} = 0.642$) indicate that this divergence is not statistically significant at the 95% confidence level, constrained by the reduced sample size of the isolated temporal panel. This yields the definitive answer to RQ2: the fundamental "rules" and penalties for financial and energy deprivation did not statistically decouple or change. Rather, the *predictability* of happiness based on these material factors nearly doubled. The global economy did not invent new rules in 2023; it simply began enforcing the existing penalties of material deprivation with absolute strictness.

```

=== POOLED INTERACTION MODEL (STATISTICAL SIGNIFICANCE) ===
=====

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.1390	0.312	-0.445	0.656	-0.751	0.473
Inability_to_keep_home_adequately_warm	-0.4383	0.392	-1.118	0.264	-1.207	0.330
is_2023	-0.0007	0.312	-0.002	0.998	-0.612	0.611
Inability_to_keep_home_adequately_warm:is_2023	-0.2223	0.477	-0.466	0.642	-1.158	0.714
Employment_rate	0.0147	0.539	0.027	0.978	-1.042	1.071
Employment_rate:is_2023	0.0865	0.623	0.139	0.890	-1.135	1.308
Feelings_of_physical_pain	0.4142	0.249	1.666	0.096	-0.073	0.901
Feelings_of_physical_pain:is_2023	-0.0396	0.356	-0.111	0.912	-0.738	0.659
Difficulty_making_ends_meet	-0.2031	0.540	-0.376	0.707	-1.261	0.855
Difficulty_making_ends_meet:is_2023	-0.0024	0.488	-0.005	0.996	-0.959	0.954
I(Inability_to_keep_home_adequately_warm ** 2)	0.0961	0.248	0.388	0.698	-0.389	0.582
I(Difficulty_making_ends_meet ** 2)	0.0507	0.099	0.509	0.611	-0.144	0.246

```

=====

```

Figure 12: Pooled Interaction Model. Regression results testing for temporal shifts. The high p -values across the interaction terms (e.g., $p\text{-value} > 0.05$) mathematically confirm that the fundamental baseline penalties for material deprivation did not structurally decouple or change between 2021 and 2023.

The comparative visualization of standardized OLS coefficients between the 2021 pandemic crisis and the 2023 inflationary recovery illustrates a profound structural consolidation of macroeconomic penalties. Most notably, the psychological penalty for energy poverty (Inability to keep home adequately warm) drastically intensified in 2023, reflecting the severe toll of the post-pandemic global energy crisis on national well-being [4]. Conversely, the negative impact of general financial stress (Difficulty making ends meet) remained visually and statistically constant across both periods, proving that the fundamental psychological burden of baseline poverty did not structurally decouple. Finally, the chart highlights the re-emergence of labor stability: while the Employment rate was statistically mute during the erratic 2021 lockdowns, it returned as a distinct positive driver of life satisfaction in 2023 as pandemic-era labor anomalies subsided and traditional economic rules strictly reasserted themselves.

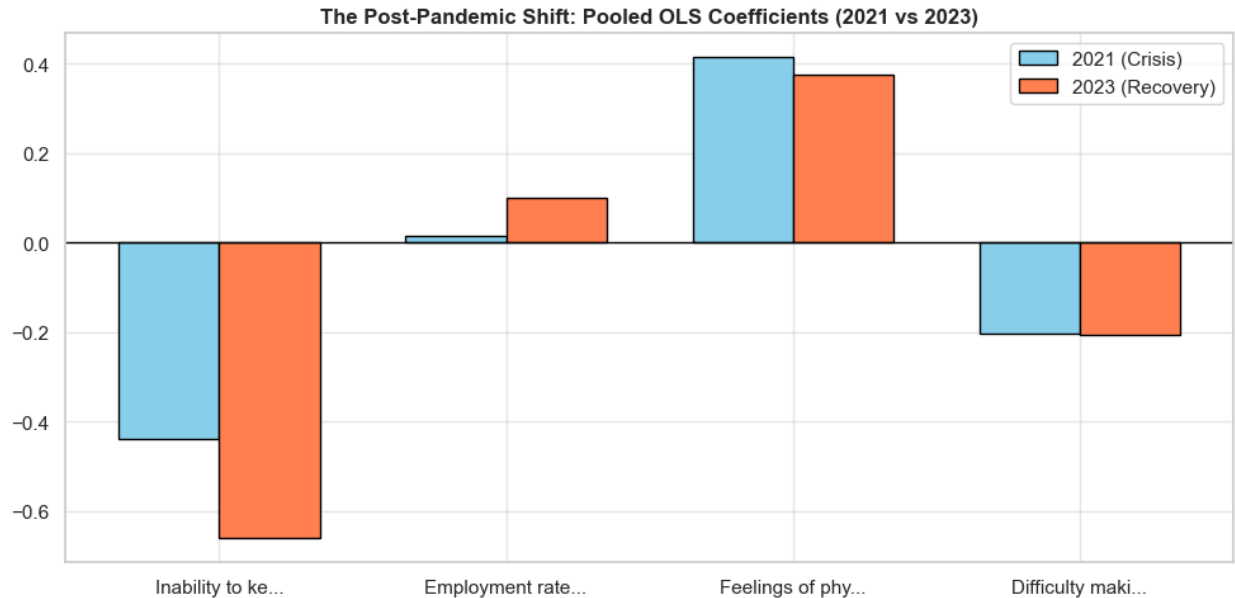


Figure 13: Post-Pandemic Shift in Macroeconomic Penalties. A comparative visualization of standardized OLS coefficients. The graph visually confirms that while the penalty for general financial stress remained constant (far right), the psychological penalty for energy poverty drastically worsened during the 2023 inflationary recovery (far left). Furthermore, baseline employment re-emerged as a positive driver of well-being as pandemic-era labor anomalies subsided.

8. Conclusion

8.1. The Gap Between Initial Objectives and Final Results

The initial objective of this research was to leverage advanced Machine Learning to independently extract and predict the universal drivers of global well-being. However, the empirical results revealed a profound structural gap between pure algorithmic prediction and macroeconomic reality. While the Random Forest architecture successfully achieved high initial accuracy ($R^2 = 0.521$), rigorous cross-validation exposed severe out-of-sample instability (Mean $R^2 = -1.117$). The unconstrained algorithms struggled to generalize across the high-variance, low-sample environment inherent to sovereign datasets.

Rather than viewing this gap as a failure, this research utilized it as a critical finding. The realization that tree-based ensembles are structurally vulnerable to macroeconomic outliers necessitated the development of a Dual-Model Strategy. Explainable AI (SHAP) was repositioned from a standalone predictor to an exploratory tool, utilized exclusively to map non-linear thresholds—such as the massive psychological penalty of extreme financial stress. These algorithmic discoveries were then passed to classical HC3 OLS econometrics for rigorous causal validation. Ultimately, bridging this gap between AI exploration and econometric validation became the defining methodological achievement of the thesis.

8.2. Strengths and Weaknesses of the Study

The primary strength of this study lies in its synthesis of modern Explainable AI with traditional econometric rigor. By utilizing SHAP values, the research successfully visualized non-linear socioeconomic thresholds that traditional linear models would have ignored. Furthermore, the methodology successfully identified and neutralized severe confounding variables—most notably the "Demographic Paradox," where pure predictive AI mistakenly correlated high physical pain with high life satisfaction due to the aging populations of wealthy, developed nations. Finally, the temporal segmentation provided empirical, mathematical proof of the post-pandemic economic consolidation, demonstrating how the predictability of national happiness nearly doubled between the erratic 2021 crisis and the strict 2023 recovery.

Despite these strengths, the study acknowledges distinct limitations, primarily stemming from data sparsity. Macroeconomic analysis relies on sovereign national averages, restricting the Golden Dataset to 153 total observations. When segmenting the data temporally for the post-pandemic analysis, the sample size was reduced to approximately 30 nations per year. This limitation directly constrained the statistical power of the Pooled Interaction Model; while visual coefficient shifts indicated a worsening penalty for energy poverty in 2023, the reduced sample size artificially inflated the p-values, preventing the confirmation of a statistically significant slope divergence at the 95% confidence level.

8.3. Directions for Future Research

To bypass the limitations of sovereign-level data sparsity, future research must pivot from macroeconomic averages to microeconomic census data. Applying this exact Dual-Model Strategy (Random Forest feature extraction mapped to robust OLS) to individual-level household surveys containing millions of data points would eliminate the cross-validation instability observed in this study. A massive sample size would allow unconstrained Machine Learning algorithms to identify highly granular, localized thresholds for energy poverty and financial stress without the risk of overfitting national outliers.

Additionally, future iterations should incorporate rigorous time-series forecasting. While this study utilized cross-sectional temporal segmentation (isolating 2021 versus 2023), applying Panel Data econometrics—such as Fixed Effects or Autoregressive Distributed Lag (ARDL) models—would allow researchers to measure the *lagged* psychological effects of material deprivation. Understanding exactly how many months it takes for a spike in energy poverty to permanently depress national life satisfaction will be crucial for policymakers attempting to deploy proactive, rather than reactive, economic interventions.

9. References

- [1] (2026) OECD Data Explorer. [Online]. [Link](#)
- [2] J. F., Layard, R., Sachs, J. D., De Neve, J.-E., Akinin, L. B., & Wang, S. (Eds.) Helliwell, "World Happiness Report. Sustainable Development Solutions Network," 2024.

- [3] A. Deaton, "Income, health, and well-being around the world: Evidence from the Gallup," 22(2), 53-72., 2008.
- [4] H., Snell, C., & Bouzarovski, S. Thomson, "Health, well-being and energy poverty in Europe: A comparative study of 32 European countries. International Journal of Environmental Research and Public Health, 14(6), 584.," 2017.